L1. Methodology

Method

What it is?

A method is an empirical practice used in science to produce knowledge. Methods are concrete ways of doing research such as observations experiments or model studies. A method is a tool for collecting evidence or testing claims. A method alone does not explain why it was chosen.

Example(s)

- Observing traffic flow at an intersection
- Running a laboratory experiment on materials
- Using a computer model to simulate weather

Methodology

What it is?

Methodology is the systematic assessment and justification of method choice. It focuses on why one method is chosen instead of another. Methodology depends on the scientific goal and the situation in which the research is done.

Example(s)

- Arguing why a field study is better than a lab experiment for a research question
- Justifying the use of simulations because experiments are not possible

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Conventional view of methodology

What it is?

The conventional view says that scientists choose methods because they are commonly used in their field. The method choice is based on

tradition rather than critical evaluation.

Example(s)

- Always using statistical tests because everyone in the field does
- Following a supervisor's preferred method without questioning it

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Best results view of methodology

What it is?

The best results view says that scientists should choose the method that gives the best results. This view is unclear because it does not explain what best means or how to judge results in advance.

Example(s)

- Choosing a method because it produces results quickly
- Defending a method by claiming it might lead to future benefits

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Epistemic tool view of methodology

What it is?

The epistemic tool view says that methods are tools chosen because we have good reasons to believe they will achieve a scientific goal. Method choice is based on reasons given before the method is used.

Example(s)

- Choosing experiments to find causes
- Using observational studies when experiments are not allowed

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Prediction

What it is?

Prediction is a scientific goal that involves knowing that something will happen in the future. Predictions are often uncertain.

Example(s)

- Predicting the weather
- Predicting how a disease will spread

Explanation

What it is?

Explanation means knowing what caused an event or outcome. An explanation aims to identify causes not just describe patterns.

Example(s)

- Explaining a crash by identifying brake failure
- Explaining climate change by identifying human activity

Design

What it is?

Design is a scientific goal focused on creating objects or systems that perform certain functions. It involves knowing how to make something work as intended.

Example(s)

- Designing a safe bridge
- Creating a medical device

Belief

What it is?

A belief is a mental state where a person thinks that a claim is true. Beliefs can be strong or weak and do not require certainty.

Example(s)

- Believing a theory explains a phenomenon
- Believing a coin toss will land heads

Justification

What it is?

Justification is the evidence or reasons used to defend a belief. In science justification must be possible to explain and defend to others.

Example(s)

- Using data to support a conclusion
- Referring to experiments to defend a claim

Truth

What it is?

Truth means that a claim correctly describes how things really are. A belief can be justified but still false.

Example(s)

- The Earth orbits the Sun
- The geocentric model was once justified but false

Uncertainty

What it is?

Uncertainty means that a belief is not fully certain. Scientific knowledge often includes uncertainty.

Example(s)

- Estimating risk with probabilities
- Making predictions with margins of error

Instrumental reasoning

What it is?

Instrumental reasoning is reasoning about how to achieve a goal. It involves choosing the best means to reach that goal.

Example(s)

- Choosing a method that best fits a research goal
- Selecting simulations when real testing is not possible

Instrumentalism

What it is?

Instrumentalism is the view that a scientific theory is not strictly true or false. A theory is seen as a useful tool for organizing observations and making predictions. The entities the theory talks about are treated as conceptual tools rather than things that actually exist.

Example(s)

- Treating electrons as useful concepts for predicting experimental results without claiming that electrons really exist
- Using a model of ideal gases to make accurate predictions even if the model does not describe reality exactly
- Using economic models that help predict markets without assuming the model reflects how the world truly is

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Realism

What it is?

Realism is the view that a scientific theory can be true or false. According to realism the entities described by a theory either exist or do not exist. Scientific theories aim to describe how the world really is not just to be useful tools.

Example(s)

- Believing that electrons exist as real particles
- Treating gravitational waves as real physical phenomena
- Believing that genes are real entities that cause biological traits

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L2. Scientific Inference

Inference

What is it?

The process of reaching a conclusion from some facts or evidence.

Premise

What is it?

Premise is the starting assumption we take as true for the given problem. it's essentially a statement.

Example:

All balls are green

This object weighs 100kg

The moon is round

Different types of inferences

Direct inference

- Moves from a statement about a sample to a statement about a population.

Examples:

- **Premise**: The sky is blue **and** the grass is green → **Inference**: Therefore, the sky is blue.
- Seeing that

Generalization

What is it?

> We see several cases of something, and we conclude a general rule.

Form:

- We have seen: A, B, C are like this
- So we think: all (or most) things of this kind are like this

Example:

- You see 10 swans, and all of them are white.
- You conclude:
  "All swans in the world are white."

This is not guaranteed to be true (maybe a black swan exists), but it feels reasonable based on what weve seen.

Projection

We have seen a pattern so far, and we **project** it onto a **new case** (often in the future or somewhere else).

Form:

- So far, all Xs we've seen have property Y
- There is a new X we haven't checked yet
- We guess: the new X also has property Y

Example 1 (future):

- Every morning so far, the sun has risen in the east.
- Tomorrow is another morning.
- You conclude:
  "Tomorrow the sun will rise in the east."

Example 2 (medicine):

- A medicine helped many patients with the same illness.
- A new patient has that illness.
- You conclude:
  "This medicine will probably help this new patient too."

Projection is about applying a pattern to a new, unobserved case.

Modus Ponens (deductive)

“If P then Q. P is true. So Q must be true.”

This is a **valid deductive rule**: if the premises are true (i.e premises=statements), the conclusion must be true.

Form:

1. If P, then Q
2. P
3. Therefore, Q

Example:

1. If it is raining, then the streets are wet.
2. It is raining.
3. Therefore, the streets are wet.

If (1) and (2) are true, (3) cannot be false.

Modus tollens(deductive)

If P then Q. Q is not true. So P cannot be true. Like Modus pollens it is also deductive. But here we are doing the inversion. If P→ Q and Q is false then P cannot be true

Form:

1. If P, then Q
2. Not Q
3. Therefore, not P

Example:

1. If the fire alarm is working, then it will sound when there is smoke.
2. There is smoke, but the alarm is not sounding.
3. Therefore, the fire alarm is not working.

Again, if (1) and (2) are true, the conclusion (3) must be true.

Deductive Inferences

What it is:

Deductive inference is a type of reasoning where we start with **general statements or rules**, and from them we derive a conclusion that must be true **if the premises are true**. Deduction does **not add new information**, it simply *unpacks* what is already contained in the premises. Often times its pretty clear from the statement what will follow, its logical.

Example:

1. All humans are mortal.
2. Socrates is a human.
3. Therefore, Socrates is mortal.

Another example:

- All stars burn
- Our sun is a star
- Therefore our sun burns.

There is no way for the conclusion to be false if both premises(statements) are true. That is why deductive reasoning is described as truth-preserving.

The following are Deductive inference:

- Modus Ponens
- Modus Tollens

Inductive Inference

Inductive inference is a type of reasoning where we move from **specific observations** to a **general conclusion**. It goes beyond the information we already have, which means it can be **useful but never guaranteed to be true**. Induction is the kind of reasoning we use in everyday life, science, and prediction. Unlike deductive Inference the conclusion is *probable(might be false)*, not logically certain.

Example:

- You observe that every cat you have seen so far has whiskers.
- You conclude: "All cats have whiskers."

This may be true, but it is not logically guaranteed. One day you might see a cat without whiskers (e.g., due to injury or mutation), which would prove the generalization false. Induction is useful because we cannot observe everything, but we can still make reasonable expectations about the world.

Another example (science):

- Many experiments show that metals expand when heated.
- Scientists conclude: "All metals expand when heated."

  This is not absolutely certain, but strong evidence makes it reasonable to believe.

David Humes critique of inductive inference

Hume asks whether we can justify using induction - that is, whether we have a good reason to trust it. His answer is no, and he gives the following reasoning:

1. We want to justify inductive reasoning

Induction is something we use all the time, for example:

- "The sun has risen every day so far, so it will rise tomorrow."
- "Bread has always fed me, so it will feed me next time I eat it."

1. We cannot justify induction using deduction

   A deductive argument gives a conclusion that must be true if the premises are true.
   But inductive conclusions are not guaranteed - they go beyond what we have observed.
   So we cannot say:
   "The sun has risen every day so far, therefore it must rise tomorrow."
   Because logically, it might not. So deduction cannot save induction.

2. We also cannot justify induction by using more induction

   We might try to defend induction like this:
   "Induction has worked many times in the past, so it will work in the future."

   But this is itself an inductive argument - it assumes the future will be like the past.
   So we would just be using induction to justify induction, which is circular reasoning.

3. Therefore, induction cannot be justified

   Any attempt to defend induction either:

   - fails logically (deduction doesn't work), or

- assumes the very thing it is trying to prove (using induction to justify induction).

So Hume concludes we have no rational proof that induction is reliable, even though we all use it.

Example:

Inductive reasoning:

- Every time I have touched fire, it has burned me.
- So I believe fire will burn me next time too.

Hume's point:

- You cannot deductively prove that the fire will burn you next time - it's not logically guaranteed.
- You also cannot say "It always worked before, so it will work again," because that is already assuming that the future will be like the past - which is exactly what needs to be proven.

So the reasoning is useful, but not logically justified.

Foundationalism

Foundationalism is an argument against David Hume.

What is it?

Foundationalism is the view that knowledge must be built on **basic beliefs** that are certain, secure, and do not need further justification. These basic beliefs act like the foundation of a building: once they are in place, other beliefs can be justified by resting on them. This is used as an argument against David Hume, Foundationalists disagree with Hume because they think there can be secure basic beliefs that do not need justification from past experience.

Example:

Think of the axioms in mathematics, like 0 is a natural number. Or the shortest distance between two points is a straight line. From this we can build theories.

Coherentism

What is it?

They believe that beliefs are not justified by a single foundation, like im happy because I saw my friend. Instead they believe that several beliefs working together and support each other make the belief possible.

Instead of:

> “I am happy because I saw my friend” (just one reason)
>
> Coherentism says:
>
> “I am happy because I saw my friend, I’ve been feeling good lately, the weather is nice, and I’m not stressed.”
>
> Several beliefs together make the main belief more justified

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Example:

Hume says we cannot prove the sun will rise tomorrow, because that is just assuming "the future will be like the past."

A coherentist replies:

We don't believe the sun will rise tomorrow because of one single rule ("it rose before, so it will rise again").
We believe it because this fits with many other beliefs that support each other, such as:

- "The laws of physics"

- "The movement of the Earth"
- "Observations from astronomy"
- "Past experience"
- "Scientific models that successfully predict sunrise times"

Even if one belief failed (e.g., an old astronomy model), the others still hold, so the belief remains justified inside the network.

So Coherentism does not try to escape Hume by proving induction, but instead says Hume's demand for a single foundation is the wrong way to think about justification.

The Hypothetical deductive method (HDM)

1. Formulate a hypothesis H
2. Deduce what observable consequences (C_i) from H
3. Test whether (C_i) is true or not.
4. If (C_i) is false infer that H is false.
5. If (C_i) is true, increase confidence in $`H`$.

Why is HDM both deductive and inductive?

Deductive:

If the hypothesis is true → then we should observe X .
We did not observe X therefore the hypothesis is false.
This is modus tollens, a valid deductive rule.

So deduction is used for falsifying hypotheses.

Inductive:

If the predicted observation does happen, we cannot logically prove the hypothesis is true - we only gain support for it. That is inductive reasoning:

The prediction came true → so the hypothesis is probably right.
But it could still be false in the future, so the support is not certain.

So induction is used for confirming hypotheses.

What is a good hypothesis?

- A statement that is either true or false.
  - Example: All metals expand when heated.

- A statement that is not necessarily true or false.
  - Example : Dark matter interacts only through gravity.

- Has generality or refers to the unobservable
  - Example: Electrons have a negative charge.
  - Example: All dogs need food to survive

What is a good observable sequence $`{(C_i)}`$?

- (C_i) must be observable directly or with the help of accurate measurements (e.g. microscope, X-ray, etc.) Something we can check essentially.

- Deduction must be valid. In other words we should be able to say H is false using {C_i}

- $`{(C_i)}`$must be relevant for H.
  - **Example**:
    A hypothesis H can be that: **Plants need sunlight to survive**, the predicted consequence {C_i} can be : **If the hypothesis is true, then a plant kept in the dark will not grow well.**

Poppers Falsification

What is it?

Karl Popper argued that scientific theories can never be proven true, only proven false. A good scientific theory must make bold, testable predictions - predictions that could turn out to be wrong.

- If a prediction fails, the theory is falsified and must be rejected or revised.
- If a prediction succeeds, the theory is not proven true - it has only survived a test.

So for Popper, science progresses by eliminating false theories, not by confirming true ones.

Key idea:

Science should aim at falsifiability, not verification. A theory is scientific only if it can, in principle, be shown false.

Popper's slogan:
"One counterexample is enough."

Example:

- "All swans are white" is falsifiable - one black swan disproves it.

- "All stars are squared, our sun is round so it disproves it.

Argument against Poppers Falsification

Duhem-Quine Thesis

What is it?

> The idea is that when we test a hypothesis we never a test it alone. Instead we always rely on other assumptions, such as background theories, measurement tools, math, and auxiliary hypotheses (smaller hypothesis = auxiliary hypothesis).

So if a prediction fails, we cannot know for sure which part is wrong:

- the main hypothesis,
- one of the supporting assumptions,
- the experimental setup,
- the data interpretation, etc.

This means that we can't simply say the hypothesis failed because the experiment failed. Instead we need to look into it more carefully and see if we made the wrong assumptions, wrong setup etc.

Example:

> Suppose we test the hypothesis:
> H: "This metal expands when heated."
>
> We heat it and nothing happens. Does that mean H is false?

Not necessarily. Maybe:

- the thermometer was broken,
- the heating device malfunctioned,
- the sample wasn't actually metal,
- the temperature didn't rise enough,
- we used the wrong formula for expansion.

So the failed prediction does not automatically falsify the main hypothesis - something else in the background assumptions might be at fault.

Ad hoc modification

What is it?

Ad hoc modification is when a scientist changes or adds new assumptions to a theory only to protect it from being falsified, instead of improving the theory in a real scientific way.

Example:

Theory:
"All planets move in perfect circles."

Observation:
Mars does not move in a perfect circle.

Ad hoc fix:
"Mars is pushed by an invisible force that we cannot detect."

This saves the theory, but it does not explain anything new, predict anything useful, or make the theory stronger - it only protects it from being proven wrong.

Hypothesis confirmation

Relevance

What is it?

When we test a hypothesis, **not all observations count as good evidence**. A piece of evidence is *relevant* only if it actually makes a difference to whether the hypothesis is likely true or false

Example:

Hypothesis: "This medicine reduces fever."

- Relevant evidence: Patients who take the medicine have lower temperature than those who don't.
- Irrelevant evidence: The pills are blue, the doctor likes the medicine, the patients slept well.

So relevance means: the evidence must be connected to what the hypothesis claims.

Underdetermination

What is it?

Underdetermination means that the same evidence can fit several different hypothesis, so the data by itself cannot tell us which explanation is the right one.

Example:

Data: The population of a city increased every year for 10 years.
Possible hypotheses:

1. "The birth rate is rising."
2. "More people are moving into the city."
3. "Death rates have fallen due to better healthcare."
4. "A combination of all three."

The same data fits multiple explanations - so the hypothesis is underdetermined by the evidence.

L3. Observation and Measurement

Empiricism and Types of Observation

Empiricism

- **Empiricism**: the view that **sensory experience** is the ultimate source of knowledge.
- In the end, scientific claims must connect back to **what we can observe** with our senses, sometimes helped by instruments.

Direct Observation

- You use **only your own senses** to experience the event or object.
- **Examples:**
  - Seeing that a car is moving and not standing still.
  - Feeling that one bag is heavier than another.
  - Hearing that one sound is louder than another.
- **Limitations of direct observation**:
  - It can be **mistaken**:
    - A color blind person can misjudge colors.
    - Strange lighting can change how colors or shapes appear.
  - It is usually **qualitative**, not numerical:

    - You can tell "this is warmer than that", but not exactly *how much* warmer.
    - You can tell "this bag feels heavier", but not its exact weight in kilograms.

Aided Direct Observation

- You still perceive the **thing itself**, but with help from an instrument.
- **Examples**:
    - Glasses or contact lenses to see more sharply.
    - A microscope to see cells or bacteria.
    - A telescope to see distant planets or moons.
- These observations often depend on **background theory**:
    - To trust a telescope, we rely on a theory of **optics** that tells us how lenses bend light.
    - In Galileo's time, some scholars thought telescopes produced illusions, because they did not yet trust the theory.
- **Key idea:**
    - Aided direct observation is still "looking at the object", but the reliability of what you see depends on a theory of how the instrument works.

Indirect Observation

- You do **not** observe the phenomenon itself.

  Instead, you observe an **effect** that the phenomenon causes.
- General structure:

  > Phenomenon (hidden or hard to observe)
  >
  > → causes some effect
  >
  > → we observe the effect and infer the phenomenon
- Examples:
    - We do not see **gravity**, but we see planets move in orbits and objects fall.

    - We do not see **elementary particles**, but we see **tracks** in a bubble chamber.
    - We do not see **temperature itself**, but we see mercury rise in a thermometer.
- **Indirect observations are always theory dependent**:
    - We need a theory that explains how the phenomenon produces the observable effect.
    - Without that theory, the effect would just be "bubbles in liquid" or "lines on a screen" and would not count as evidence for particles or fields.

Theory Dependence of Observation

- An observation is **theory dependent** if its **interpretation** depends on some accepted theory.
- Example with a microscope:
    - A scientist says "I see bacteria".
    - Strictly speaking, they see tiny shapes or dots in a microscope image.
    - It is the **theory of optics** plus **cell theory** and **staining techniques** that justifies calling those dots "bacteria".
- This shows:
    - Many observations, especially aided and indirect ones, are **not neutral**.
    - They already presuppose some theoretical assumptions in order to count as observations *of* something specific.

Logical Empiricism and Circularity

Logical Empiricism

- **Logical empiricism** is a philosophical movement with roughly these ideas:
    - Meaningful statements are either:
        - **empirically testable**, or
        - **logically or mathematically true**.

    - Metaphysical statements that cannot be tested are seen as **meaningless** for science.
    - Philosophy should clarify scientific language and logic, not introduce new untestable claims.
- Examples of how a logical empiricist classify statements:
    - "Electrons have negative charge" → meaningful, because it is testable in principle.
    - "God exists beyond space and time" → not meaningful for science, because it is not testable.
    - "2 + 2 = 4" → meaningful, because it is analytically or logically true.
- Problem highlighted in the chapter:
    - Logical empiricism needs a **clear distinction** between:
        - observational sentences, and
        - theoretical sentences.
    - If many observations are **theory dependent**, this clear separation becomes difficult or impossible.

Threat of Circularity in Indirect Observation

- In indirect observation, we:
    1. Use a **theory** to interpret what we see.
    2. Then use that interpreted observation as **support** for the same theory.
- This risks a kind of **circularity**.

Example from particle physics:

- We see **bubble tracks** in a chamber.
- Theory says:
    - particles cause ionization in the liquid,
    - ionization causes bubbles,
    - the shape of the track is related to charge, mass, and momentum of the particle.
- So we say "We observed a particle with these properties".

- But how do we know the theory about ionization and track shapes is correct?
  - Because it fits the bubble observations.

So:

- The **theory** is used to interpret the observation.
- Then the **interpreted observation** is used to support the theory.
- This is the **threat of circularity** in indirect observation.

The main lesson:

Observation and theory support each other, rather than having a simple one way order where observation comes first and theory comes later.

Operationalization and Operationalism

Operationalization

- **Operationalization** is the process of turning an **abstract or unobservable property** into something we can **measure**.
- Examples of properties of interest:
  - Temperature
  - Stress
  - Intelligence
  - Political attitude
  - Acidity
- Examples of operationalizations:
  - Temperature → height of mercury in a thermometer.
  - Stress → self report questionnaire and cortisol level in saliva.
  - Intelligence → score on an IQ test.
  - Acidity → color of pH paper.
- Basic structure of an operationalization:
  1. **Property of interest**

- What we want to measure, for example temperature or happiness.

2. **Hypothesized causal or correlational link**
   - A theory that says changes in the property lead to changes in some observable effect.
     Example: higher temperature leads to more expansion of a liquid in a thermometer.
3. **Directly observable effect**
   - What we actually look at, for example the height of a liquid column, the color of a strip, or answers on a survey.

A **good** operationalization has:

- A **clear property of interest**.
- A **well justified and sufficiently stable relationship** between the property and the effect.
- An **effect that is easy enough to observe with sufficient precision**.

Operationalism (too strong)

- **Operationalism** is a stronger and more problematic claim:
  - It says that a scientific concept **is nothing more than** the operation or measurement used to detect it.
- Examples of operationalism:
  - "Temperature is what the thermometer reads."
  - "Intelligence is the IQ test score."
  - "Happiness is the number you give on a 0 to 10 scale."
- Problems with operationalism:
  1. **Concept understanding without simple measurement**
     - We can understand concepts even when measurement is very complex or not yet possible, for example extremely small lengths or distances of light years.
  2. **Multiple measurements for the same concept**

    - Different thermometers or sensors can all measure temperature.
      If operationalism is correct, each method would define a different "temperature", which is not how we use the concept in practice.
3. **We want to criticize measurements**
    - In real science, we often ask "Is this test a good measure of intelligence?"
      If "intelligence = IQ score", such criticism makes no sense.

- Conclusion:
    - **Operationalization** is necessary and good.
      It is about making concepts measurable.
    - **Operationalism** is too extreme and is rejected.
      It wrongly identifies the concept with a single measurement method.

Measurement: Steps, Units, and Scales

Steps of Measurement

To measure a property, we typically go through four steps:

1. **Define the concept to be measured**
    - Be clear about what exactly you want to measure.
    - Examples:
        - Temperature = degree of thermal energy of a system.
        - Stress = level of physiological and psychological strain.
        - Intelligence = ability to reason, solve problems, and learn.
2. **Specify the measurement operation**
    - Decide how the property will be detected in practice.
    - Examples:
        - Acidity → use pH paper and see the color change.
        - Stress → use a questionnaire plus cortisol analysis.
        - Length → place a ruler next to the object and see where it ends.
3. **Specify a measure and a unit**

- Choose the instrument and the unit or scale.
- Examples:
  - Temperature → thermometer, unit: Celsius, Fahrenheit, or Kelvin.
  - Weight → scale, unit: kilogram or pound.
  - Time → clock, unit: seconds, minutes.

4. **Represent results with numbers**
   - Record the outcome as a number, for example:
     - 37.2 °C
     - 1.5 kg
     - pH 4
     - stress score 32 out of 40

Requirements on a Good Measure

Two important criteria:

1. **Comparability**
   - The measure must have the **same type of property** as the thing being measured.
   - Examples:
     - A ruler has length. It is comparable to the length of a table.
     - A volume measure (like a measuring cup) has volume. It is comparable to the volume of a liquid.
   - A thermometer cannot directly measure length, because temperature and length are different types of properties.
2. **Stability**
   - The unit must be **sufficiently stable** over time and under normal conditions.
   - Examples:
     - A block of ice is not a stable unit of volume, because it melts and changes shape its not also satisfy comparability as ice and liquid water are not the same.

    - The oscillation of a cesium atom is extremely stable, which is why it is used to define the second.

It is also important that the measurement procedure is **public and repeatable**, so that other people can reproduce the result.

Scale Types

A **scale type** describes what kind of information the numbers carry and which mathematical operations make sense for those numbers.

Another way to define a scale type is by the **transformations of the numbers that leave the meaning of the measurement unchanged**.

Nominal Scale

- **What it is**:
    - Pure **categorization**. The numbers or labels just name different groups.
    - There is no natural order.
- **Examples**:
    - Student ID numbers.
    - Zip codes.
    - Shirt colors (red, blue, green).
    - Blood types, ethnic groups.
- **Allowed transformations:**
    - Any relabeling that preserves **non identity**.
    - For example, if dog = 1 and cat = 2, you can switch to dog = A and cat = B.
      The structure is unchanged as long as different categories stay different.

Ordinal Scale

- **What it is:**
    - A scale that represents **order** or **ranking**.
    - We know which is more or less, but not by how much.

- **Examples**:
  - Race positions: 1st, 2nd, 3rd.
  - TERRIBLE, BAD, NEUTRAL, GOOD, GREAT.
  - Movie ratings with stars.
  - Mohs hardness scale.
- We cannot say that the distance between "BAD" and "NEUTRAL" is equal to the distance between "NEUTRAL" and "GOOD".
- **Allowed transformations:**
  - Any **positive monotone** transformation, which keeps the order.
  - **Example**:

    if ranks are 1, 2, 3, we can transform them to 10, 20, 30 as long as $10 < 20 < 30$.

Interval Scale

- **What it is:**
  - A scale where **equal numerical differences** represent equal differences in the property.
  - Zero is **not a true absence** of the property. It is chosen by convention.
- **Examples:**
  - Temperature in Celsius or Fahrenheit.
  - Calendar years (1000, 1100, 1200).
- We can meaningfully talk about **differences**:
  - 20 °C is 10 degrees warmer than 10 °C.
- We cannot say:
  - 20 °C is twice as hot as 10 °C, because 0 °C is not "no temperature".
- **Allowed transformations:**
  - **Positive linear** transformations:
    $x' = a\,x + b$, with $a > 0$.

- Example: Fahrenheit and Celsius:
  $F = 1.8\ C + 32$.

Ratio Scale

- **What it is:**
  - A scale with equal intervals and a **true zero**.
  - Zero corresponds to a complete absence of the property.
  - Ratios are meaningful.
- **Examples**:
  - Length in meters.
  - Weight in kilograms.
  - Age in years.
  - Temperature in Kelvin.
- **We can say:**
  - 2 meters is twice as long as 1 meter.
  - 80 K is twice the absolute temperature of 40 K.
- **Allowed transformations:**
  - **Positive scalar** transformations:
    $x' = a\ x$, with $a > 0$.
  - Example: meters to inches:
    $\text{inches} = 39.37 \times \text{meters}$.

Absolute Scale

- **What it is:**
  - A scale with a **natural zero** and a **natural unit** that is fixed by the world, not defined by convention.
- **Examples**:
  - Counting objects: 0, 1, 2, 3 apples.
  - Probability: from 0 to 1.

- We cannot rescale these values meaningfully.
  If you have 3 apples, you simply have 3 apples.
- **Allowed transformations:**
  - No non trivial transformations are allowed.
    Any change of the numbers changes the meaning.

Scale Type and Transformations: Summary

- **Nominal**: any transformation that preserves which items are equal or different.
- **Ordinal**: any transformation that preserves the order.
- **Interval**: positive linear transformations (x' = a x + b).
- **Ratio**: positive scalar transformations (x' = a x).
- **Absolute**: no non trivial transformations allowed.

Error, Accuracy, Precision, and Validity

Systematic Error (Bias)

- A **systematic error** is a **consistent, repeatable** error that shifts all measurements in the **same direction**.
- It affects **accuracy**, that is how close the average measurement is to the true value.
- **Examples**:
  - A scale that is miscalibrated and always reads +2 kg too high.
  - A ruler that is slightly too short, so all length measurements are too small.
- A series of measurements can be:
  - **Precise but biased**.
    The results are close to each other but far from the true value.

Random Error (Noise)

- A **random error** causes measurements to **fluctuate unpredictably** around the true value.
- It affects **precision**, that is how tightly repeated measurements cluster together.
- **Example**:
  - You weigh yourself several times and get slightly different numbers each time, even though your real weight is not changing.
- **Important property:**
  - If there is **no systematic error**, the average of many repeated measurements can reduce the impact of random error and get closer to the true value.

Precision vs Accuracy

- **Precision**:
  - Low random error.
  - Repeated measurements cluster tightly.
- **Accuracy**:
  - Low systematic error.
  - Measurements are close to the true value.
- You can have:
  - High precision and low accuracy (clustered but biased).
  - Low precision and reasonable accuracy on average (values scattered around the true value).

Convergent and Divergent Validity

These concepts help us judge whether a measurement method is working properly, and they can also hint at **systematic errors**.

Convergent Validity

- **Convergent validity** is achieved when **different, independent methods** for measuring the **same property** give **similar results**.
- **Example**:
  - Three different thermometers (mercury, gas, infrared) give nearly the same temperature for the same object.
- If they do **not** converge, at least one method may contain **systematic error**.

Divergent Validity

- **Divergent validity** is defined as follows:
  A method has divergent validity if it gives **different results** when applied to **substantially different targets i**f we apply a test on A and another one to B and A and B are **different** then we should observe **different measurements**.
- **Example:**
  - A thermometer should give a low value for and ice cube and a high value for boiling water. If it doesn't, there is an issue because obviously the ice cube is colder.
  - If we measure someones heartbeat when they are running and when they are resting, we should get different results.
- If a measurement method gives almost the **same result for clearly different cases**, it may be insensitive or systematically flawed.
